

UFFT — Physical Observables from Foam Geometry

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Core Algebraic Object

The face Laplacian of the truncated octahedron has characteristic polynomial containing the master equation:

$$\lambda^2 - C_A^2 \lambda + (C_A + 1)^2 = 0$$

with $C_A = 3$ (dimension of T_2g irrep, fixed by O_h representation theory).

Roots: $r_1 = (9 - \sqrt{17})/2 \approx 2.438$ and $r_2 = (9 + \sqrt{17})/2 \approx 6.562$

This single quadratic plus the cell's topological integers generates all results below.

Cell Integers

Symbol	Value	Meaning
C_A	3	Colours / generations / spatial dimensions / T_2g dimension
$C_A + 1$	4	Proton topology (3 quarks + Y-junction) / spacetime dimensions
C_A^2	9	$\lambda_{\text{max}} / \beta_0(\text{QCD}, n_f = 3)$
$C_A^2 - 1$	8	Gluon count (adjoint dimension)
$ O_h $	48	Symmetry group order
V, E, F	24, 36, 14	Vertices, edges, faces of truncated octahedron
χ	2	Euler characteristic
d	3	Spatial dimension

Fundamental Constants

Observable	UFFT formula	Value	Experiment	Accuracy
Fine structure constant	$\alpha^{-1} = 8\pi^{5/2} \times [(G -1)/ G + (V-F)/(d \cdot G ^3) + (E-F)/(d \cdot G)]$	137.035999055	137.035999046(27) [Cs]	0.3σ

Observable	UFFT formula	Value	Experiment	Accuracy
	$\left.{}^5\right)]^{-1}$			
Dark matter ratio	$\Omega_{\text{DM}}/\Omega_{\text{b}} = 3(1+2\sqrt{3})/2^{4/3}$	5.3147	5.36 ± 0.05 [Planck]	0.85%
Dark energy density	$\rho_{\Lambda} = \rho_0(1_P/R_U)^2 \times (F-\chi)/F$	5.96×10^{-27} kg/m ³	5.88×10^{-27} [Planck]	1.4%
Speed of light	$c = \sqrt{(P_0/\rho_0)}$	Derived	Defined	Exact
Spatial curvature	$k = 0$	0	$ \Omega_k < 0.002$	Consistent

Gravity and Spacetime

Observable	UFFT formula	Status
Newtonian gravity	$g = -\nabla(c^2\rho/\rho_0) \rightarrow F = GMm/r^2$	Derived
g_{tt}	$-c^2(1 - 2GM/rc^2)$	Derived (Part XVII)
g_{rr}	$(1 - 2GM/rc^2)^{-1}$	Derived ($\nu = 1/2$ incompressibility)
Einstein equations	$R_{\mu\nu} - \frac{1}{4}g_{\mu\nu}R = (8\pi G/c^4)(T_{\mu\nu} - \frac{1}{4}g_{\mu\nu}T)$	Derived (unimodular, trace-free)
Cosmological constant	Integration constant from trace-free structure	Resolved
Friedmann equations	Both derived from foam energy conservation	Derived
Hawking temperature	$T_H = \hbar c^3/(8\pi GMk_B)$	Consistent (standard derivation in foam language)
GW speed	$c_{\text{GW}} = c_{\text{EM}} = c$	Confirmed to 10^{-15} (GW170817)

Electromagnetism

Equation	UFFT derivation route	Status
$\nabla\cdot E = \rho_e/\epsilon_0$	Volterra's theorem + Laplace in 3D	Derived

Equation	UFFT derivation route	Status
$\nabla \cdot \mathbf{B} = 0$	$\nabla \cdot (\nabla \times \mathbf{A}) \equiv 0$	Derived (identity)
$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$	$\square \mathbf{D} = 0$ + Helmholtz	Derived
$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t$	$\square \mathbf{D} = 0$ + charge conservation	Derived
Gauge invariance (U(1))	Foam translational symmetry	Derived
Charge quantisation	O_h lattice rotation: $\theta_{\min} = 2\pi / C_A$	Derived
Charge conservation	Winding number conservation = Axiom Zero	Derived
Coulomb $1/r^2$ law	Unique solution to $\nabla^2 \varphi = 0$ in 3D	Derived

QCD and Confinement

Observable	UFFT formula	Value	Experiment	Accuracy
Colour number	$C_A = \dim(T_2 g)$	3	3	Exact
Gluon count	$C_A^2 - 1$	8	8	Exact
β_0 ($n_f = C_A$)	$C_A^2 = \lambda_{\max}$	9	9	Exact
β_1 ($n_f = C_A$)	$7C_A^2 + 1$	64	64	Exact
Quark charges	$n \times 2\pi / C_A \rightarrow n \times e/3$	$e/3, 2e/3, e$	$e/3, 2e/3, e$	Exact

Light-Hadron Observables

All relations use only $C_A = 3$, the master equation integers, measured $m_u + m_d$, and observed m_p .

Observable	UFFT scaling law	Numerical (C_A = 3)	Experiment	Accuracy	Tier
Proton mass	$m_p = \pi\sqrt{2} \times \Lambda_{\text{QCD}}$	946 MeV	938.272 MeV	0.9%	Suggestive
Proton charge radius	$r_p = (C_A + 1) \times \hbar/(m_p c)$	$4\lambda_C = 0.8412$ fm	0.8414 ± 0.0019 fm	0.02%	Derived (integer)
$m_p \times r_p$ product	$(C_A + 1) \times \hbar c$	$4\hbar c$	$4.0008 \hbar c$	0.02%	Derived
$m_p^2 \langle r_p^2 \rangle / (\hbar c)^2$	$(C_A + 1)^2$	16	16.006	0.04%	Derived
Neutron $\langle r_n^2 \rangle$	$-(C_A^2 - 1)/C_A \times (\hbar c/m_n)^2$	$-8/3 \times \lambda_C^2 = -0.1176$ fm ²	-0.1161 ± 0.0022 fm ²	1.3%	Suggestive
Pion mass	$m_\pi = \sqrt{((m_u + m_d) \times C_A \times m_p)}$	139.8 MeV	139.570 MeV	0.14%	Consistent
Pion decay constant	$f_\pi = \sqrt{C_A/(C_A + 1)} \times \Lambda_{\text{QCD}}$	$\sqrt{3/4} \times \Lambda = 92.2$ MeV	92.1 MeV	0.1%	Consistent
μ_n/μ_p ratio	$-2/C_A$	$-2/3 = -0.667$	-0.685	2.7%	Suggestive
Kaon mass	$m_K = \sqrt{((m_q + m_s) \times C_A \times m_p)}$	519 MeV	493.7 MeV	5.1%	Consistent (NLO needed)

Periodic Table

Observable	UFFT formula	Value	Experiment	Status
Subshell types	O_h crystal field splitting	4 (s, p, d, f)	4	Derived
Subshell capacities	Irrep dimension $\times 2$ (spin)	2, 6, 10, 14	2, 6, 10, 14	Derived
Period lengths	Shell + subshell filling	2,8,8,18,18,32,32	2,8,8,18,18,32,32	Derived
Ordering s < p < d < f	Eigenvalue ordering $0 < r_1 < 4 < 9$	Correct	Correct	Derived
Maximum l	14 faces $\rightarrow$ 4 angular types (l = 0,1,2,3)	$l_{\text{max}} = 3$	$l_{\text{max}} = 3$	Derived

Cosmology

Observable	UFFT formula	Value	Experiment	Accuracy
Sound horizon	Standard Friedmann + foam $\Omega_{\text{DM}}/\Omega_{\text{b}}$	144.7 Mpc	144.43 ± 0.26 Mpc	0.2%
Flatness ($k = 0$)	Axiom Zero: net zero displacement	$k = 0$	$ \Omega_{\text{k}} < 0.002$	Consistent
NS sound speed	$P = \rho c^2 \rightarrow c_{\text{s}} \rightarrow c$ at high density	$c_{\text{s}} > c/\sqrt{3}$	NICER+LIGO trending	Prediction

Precision Physics Predictions (Time-Stamped, March 2026)

Prediction	UFFT value	Current data	Status
α^{-1}	137.035999055	Cs: 137.035999046(27), g-2: 137.035999206(11)	Sides with Cs (0.3σ), disagrees with g-2 (14σ)
Muon g-2 anomaly	$\Delta a_{\mu} = 19 \times 10^{-11}$ (0.3σ) with BMW HVP	Experiment: $116592059(22) \times 10^{-11}$	No new physics
Dark matter particles	None (DM = lattice geometry)	LUX/XENON null results	Prediction: null results continue
NS core c_{s}	Must exceed $c/\sqrt{3}$	NICER data trending this way	Prediction

Tier Definitions

Tier	Meaning	Count
Derived	Mathematical consequence of framework, no free parameters	8 core + periodic table + charge
Consistent	Framework accommodates correctly, doesn't uniquely predict	7 (Hawking, GW, CMB, NS, Lamb shift, etc.)
Suggestive	Pattern matches, may need additional input or NLO corrections	9+ (hadron masses, brainwaves, colour-frequency)

What Remains Open

Problem	Status	Needed
Particle mass spectrum	Knot classification programme	Major mathematical research
Kerr metric	Density prediction exists	Extend $v = 1/2$ to axisymmetric
H_0 from first principles	Boundary condition	Not derivable from dynamics
Spectral index n_s	—	Displacement cascade statistics
Baryon-to-photon ratio η	—	Foam baryogenesis
Weinberg angle running	GUT value derived (3/8)	Full particle spectrum for running
Neutron-proton mass difference	Ingredients derived	Full Cottingham calculation

The Framework in One Line

$\mathbf{B} + \mathbf{V} = \mathbf{D}$ · One axiom · One cell · One symmetry group (O_h , $|G| = 48$) · One equation of state ($P = \rho c^2$) · One master equation ($\lambda^2 - 9\lambda + 16 = 0$) · Zero free parameters · All of physics.

UFFT Core Framework v10: github.com/WebEnvy/UnifiedFoamFieldTheory Zenodo DOIs: [10.5281/zenodo.18706756](https://doi.org/10.5281/zenodo.18706756) · [18706806](https://doi.org/10.5281/zenodo.18706806) · [19011758](https://doi.org/10.5281/zenodo.19011758) · [19030062](https://doi.org/10.5281/zenodo.19030062)